Comp Eng 303 Lab 3 - Emmett Funston

SDC:

```
create_clock -name clk -period 2.0 [get_ports clk]

set_input_delay  -max 0.5  -clock clk [remove_from_collection [all_inputs] [get_ports clk]]
set_input_delay  -min -0.2 -clock clk [remove_from_collection [all_inputs] [get_ports clk]]

set_output_delay -max 0.5  -clock clk [all_outputs]
set_output_delay -min -0.2 -clock clk [all_outputs]
```

RTL:

```
`timescale 1ns/10ps

module mac_array_3x3 (
    input  wire clk,
    input  wire rstb,

    input  wire signed [7:0] IN1,
    input  wire signed [7:0] IN2,
    input  wire signed [7:0] IN3,

    input  wire signed [7:0] W1,
    input  wire signed [7:0] W2,
    input  wire signed [7:0] W3,

    output reg signed [19:0] OUT1,
    output reg signed [19:0] OUT2,
    output reg signed [19:0] OUT3,
    output reg signed [19:0] OUT4,
    output reg signed [19:0] OUT5,
    output reg signed [19:0] OUT6,
    output reg signed [19:0] OUT7,
    output reg signed [19:0] OUT8,
    output reg signed [19:0] OUT9
);

    reg signed [7:0] IN1R,  IN1RR;
    reg signed [7:0] IN2R,  IN2RR;
    reg signed [7:0] IN3R,  IN3RR;

    reg signed [7:0] W1R,   W1RR;
    reg signed [7:0] W2R,   W2RR;
```

```
reg signed [7:0] W3R,   W3RR;

reg [3:0] cycle_count;

always @(posedge clk or negedge rstb) begin
   if (!rstb) begin
      IN1R <= 8'sd0;
      IN1RR <= 8'sd0;
      IN2R <= 8'sd0;
      IN2RR <= 8'sd0;
      IN3R <= 8'sd0;
      IN3RR <= 8'sd0;

      W1R <= 8'sd0;
      W1RR <= 8'sd0;
      W2R <= 8'sd0;
      W2RR <= 8'sd0;
      W3R <= 8'sd0;
      W3RR <= 8'sd0;

      OUT1 <= 20'sd0;
      OUT2 <= 20'sd0;
      OUT3 <= 20'sd0;
      OUT4 <= 20'sd0;
      OUT5 <= 20'sd0;
      OUT6 <= 20'sd0;
      OUT7 <= 20'sd0;
      OUT8 <= 20'sd0;
      OUT9 <= 20'sd0;

      cycle_count <= 4'd0;
   end else begin

      IN1R  <= IN1;
      IN1RR <= IN1R;

      IN2R  <= IN2;
      IN2RR <= IN2R;

      IN3R  <= IN3;
      IN3RR <= IN3R;

      W1R  <= W1;
      W1RR <= W1R;
```

```
            W2R  <= W2;
            W2RR <= W2R;

            W3R  <= W3;
            W3RR <= W3R;

            if (cycle_count < 4'd13) begin
               OUT1 <= OUT1 + IN1   * W1;
               OUT2 <= OUT2 + IN1R  * W2;
               OUT3 <= OUT3 + IN1RR * W3;

               OUT4 <= OUT4 + IN2   * W1R;
               OUT5 <= OUT5 + IN2R  * W2R;
               OUT6 <= OUT6 + IN2RR * W3R;

               OUT7 <= OUT7 + IN3   * W1RR;
               OUT8 <= OUT8 + IN3R  * W2RR;
               OUT9 <= OUT9 + IN3RR * W3RR;

               cycle_count <= cycle_count + 4'd1;
            end
         end
      end

endmodule
```

Testbench:

```
`timescale 1ns/10ps

module mac_array_3x3_tb;

   reg clk;
   reg rstb;

   reg signed [7:0] IN1, IN2, IN3;
   reg signed [7:0] W1, W2, W3;

   wire signed [19:0] OUT1, OUT2, OUT3;
   wire signed [19:0] OUT4, OUT5, OUT6;
   wire signed [19:0] OUT7, OUT8, OUT9;

   mac_array_3x3 dut (
```

```
    .clk(clk),
    .rstb(rstb),

    .IN1(IN1),
    .IN2(IN2),
    .IN3(IN3),

    .W1(W1),
    .W2(W2),
    .W3(W3),

    .OUT1(OUT1),
    .OUT2(OUT2),
    .OUT3(OUT3),
    .OUT4(OUT4),
    .OUT5(OUT5),
    .OUT6(OUT6),
    .OUT7(OUT7),
    .OUT8(OUT8),
    .OUT9(OUT9)
);

always #1 clk = ~clk;

integer i;

initial begin
    clk = 0;
    rstb = 0;

    IN1 = 0;
    IN2 = 0;
    IN3 = 0;
    W1  = 0;
    W2  = 0;
    W3  = 0;

    #3;
    rstb = 1;

    for (i = 1; i <= 15; i = i + 1) begin
        @(negedge clk);

        IN1 = i;
```

```
        IN2 = i + 1;
        IN3 = i + 2;

        W1 = i + 3;
        W2 = i + 4;
        W3 = i + 5;
    end

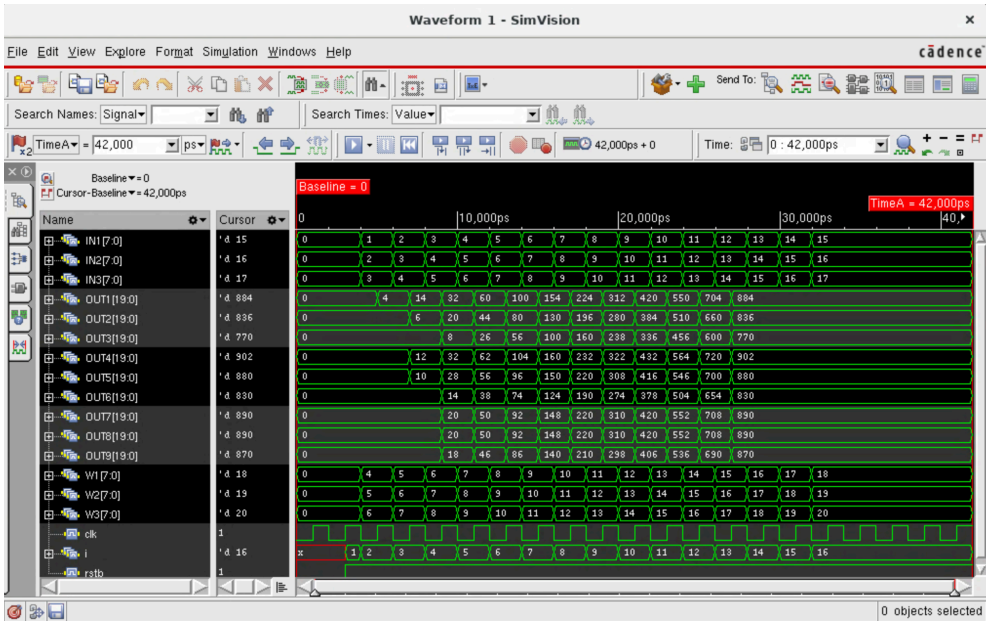
    #10;

    $display("Final outputs:");
    $display("OUT1 = %0d", OUT1);
    $display("OUT2 = %0d", OUT2);
    $display("OUT3 = %0d", OUT3);
    $display("OUT4 = %0d", OUT4);
    $display("OUT5 = %0d", OUT5);
    $display("OUT6 = %0d", OUT6);
    $display("OUT7 = %0d", OUT7);
    $display("OUT8 = %0d", OUT8);
    $display("OUT9 = %0d", OUT9);

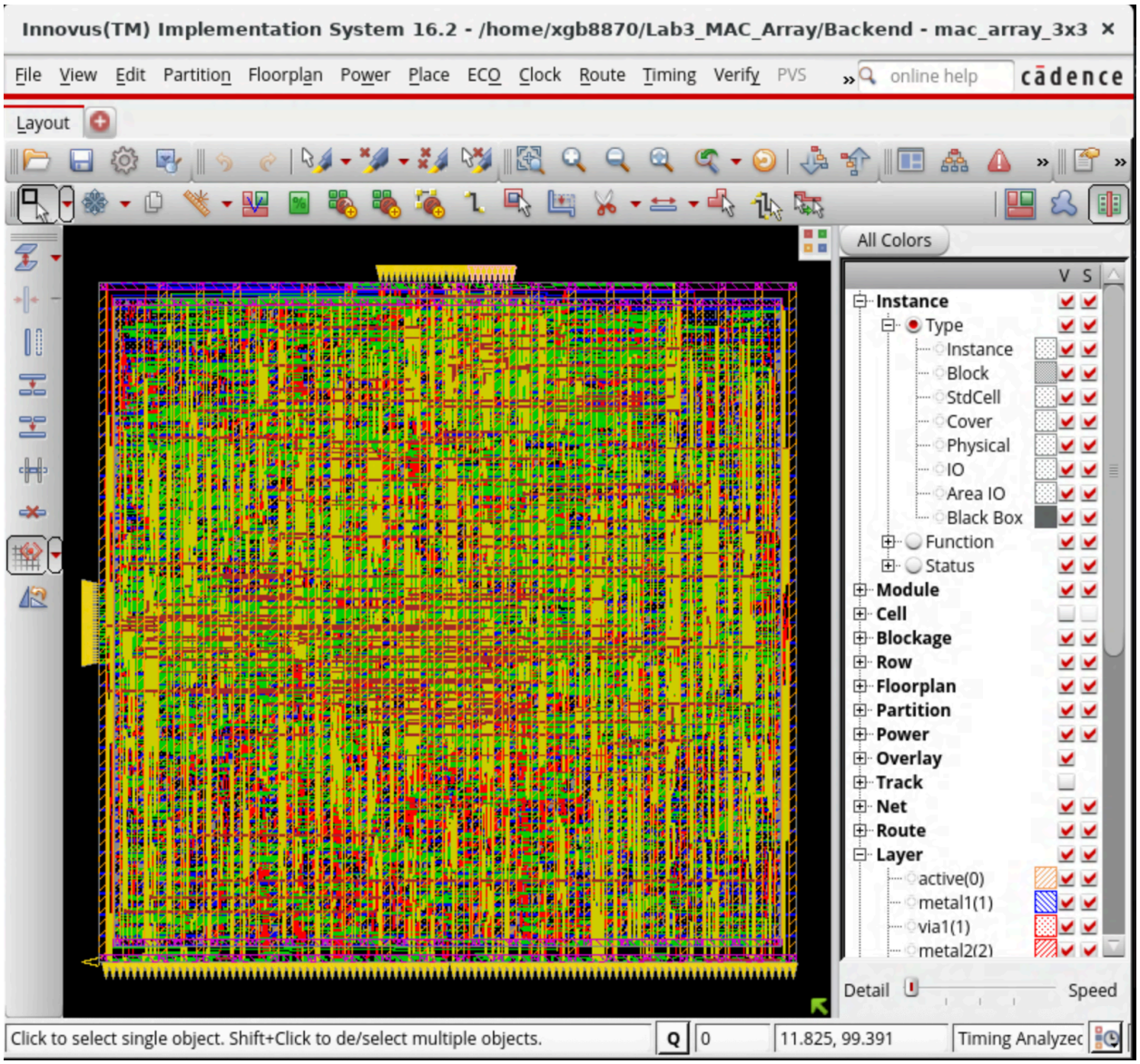
    $finish;
  end

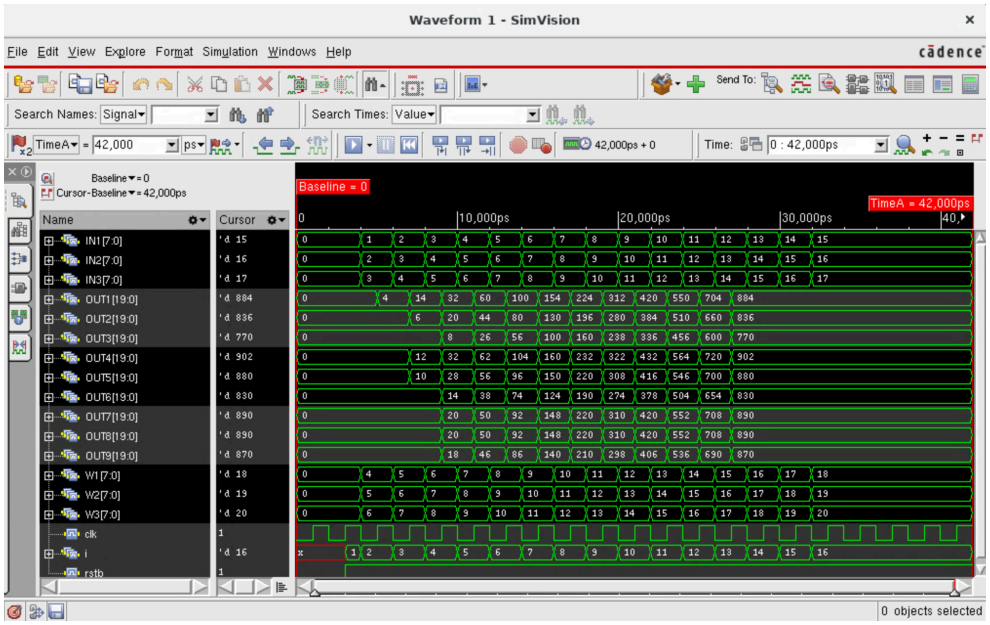
endmodule
```

Waveforms:



All are correct values.


Innovus(TM) Implementation System 16.2 - /home/xgb8870/Lab3_MAC_Array/Backend - mac_array_3x3
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS
online help
cadence
Layout
All Colors
V S
Instance
Type
Instance
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
active(0)
metal1(1)
via1(1)
metal2(2)
Detail
Speed
Click to select single object. Shift+Click to de/select multiple objects.
Q
0
11.825, 99.391
Timing Analyzer



Gatelevel matches RTL

Timing:

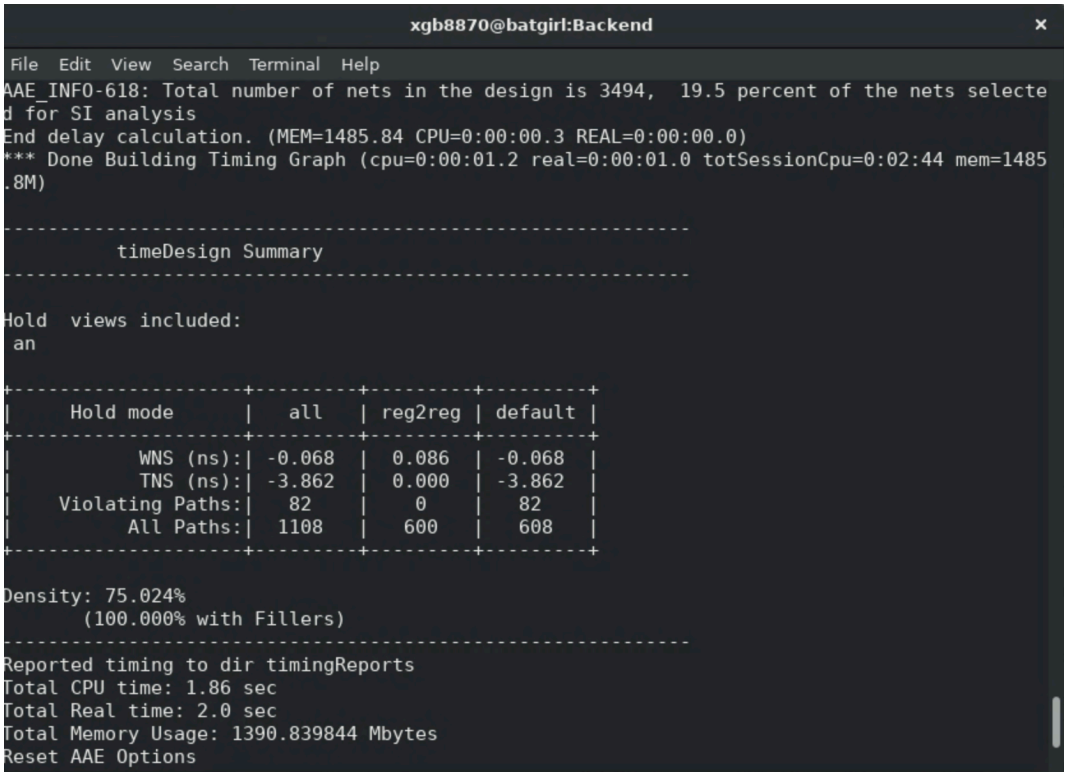
```
xgb8870@batgirl:Backend
File  Edit  View  Search  Terminal  Help
------------------------------------------------------------
         timeDesign Summary
------------------------------------------------------------

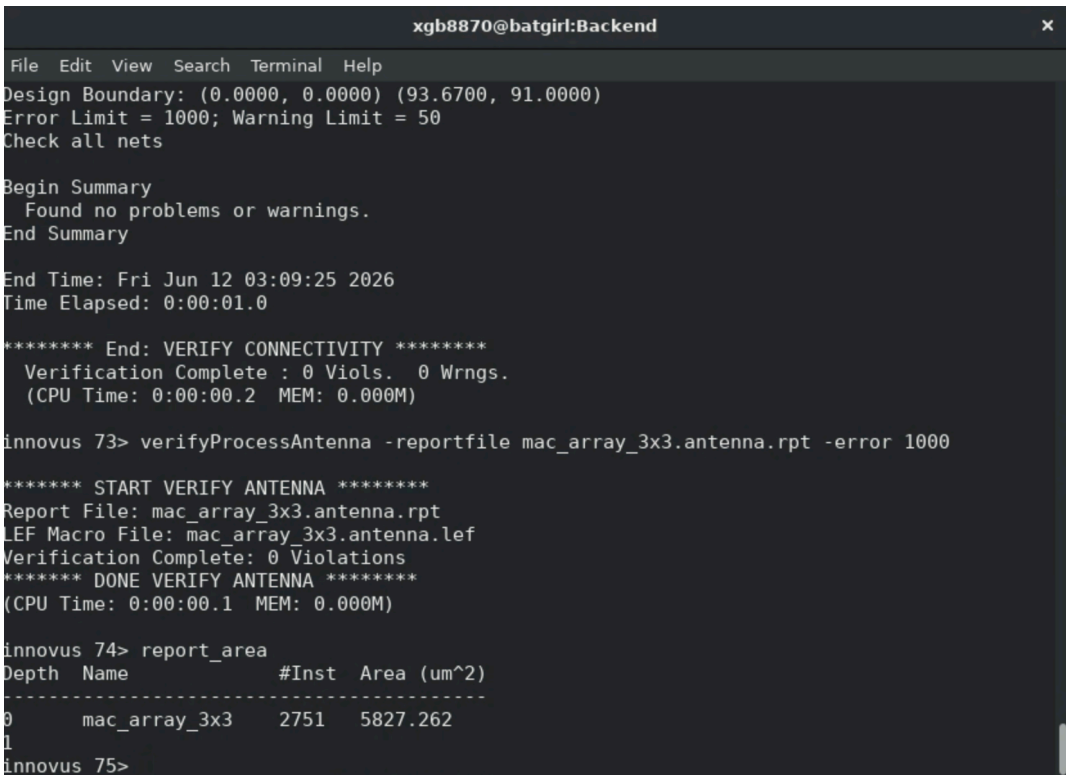
Setup views included:
 an

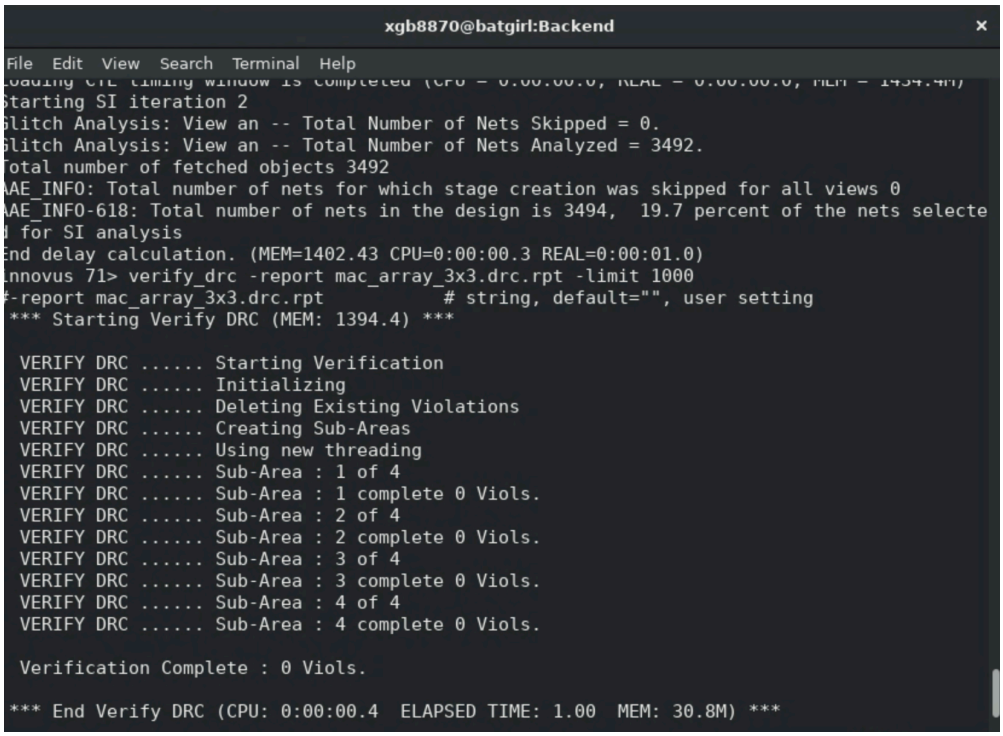
+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.115  |  0.115  |  0.115  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|  1108   |   600   |   608   |
+--------------------+---------+---------+---------+

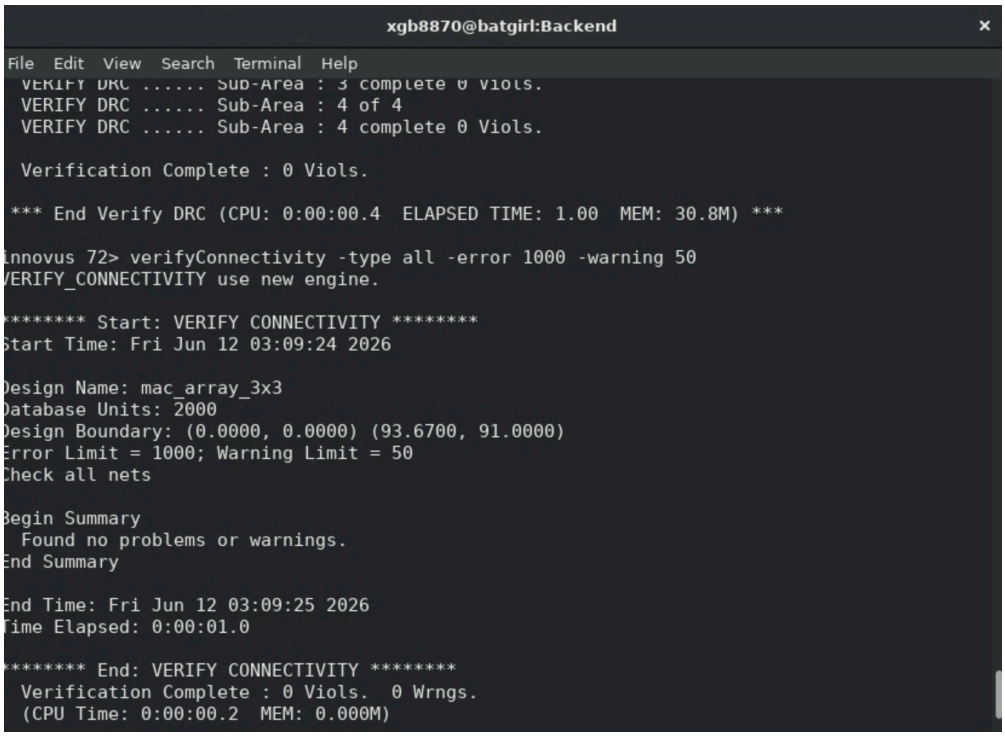
+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

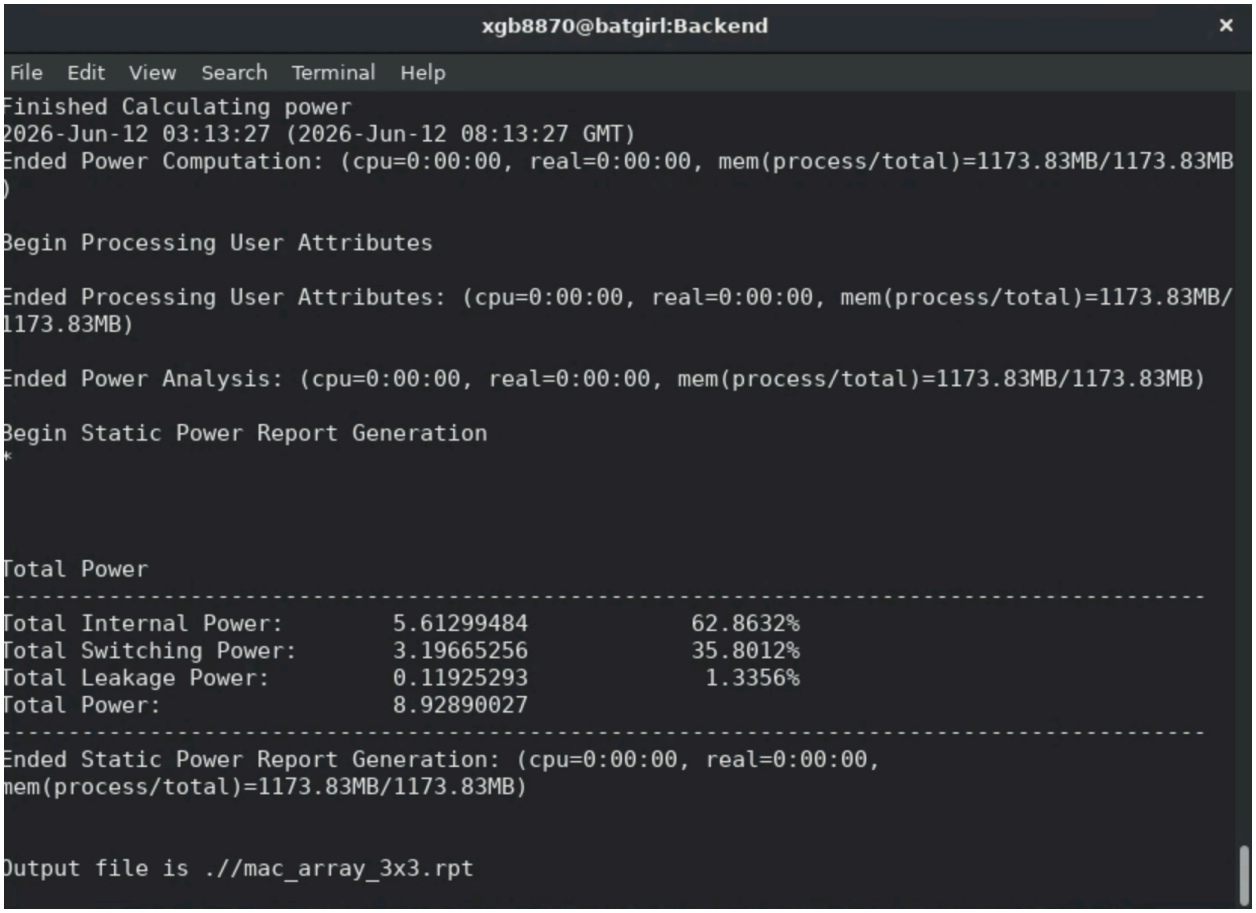
Density: 75.024%
       (100.000% with Fillers)
Total number of glitch violations: 0
```


xgb8870@batgirl:Backend
File Edit View Search Terminal Help
AAE_INFO-618: Total number of nets in the design is 3494, 19.5 percent of the nets selecte
d for SI analysis
End delay calculation. (MEM=1485.84 CPU=0:00:00.3 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:01.2 real=0:00:01.0 totSessionCpu=0:02:44 mem=1485
.8M)
timeDesign Summary
Hold views included:
an
Hold mode	all	reg2reg	default
WNS (ns):	-0.068	0.086	-0.068
TNS (ns):	-3.862	0.000	-3.862
Violating Paths:	82	0	82
All Paths:	1108	600	608
Density: 75.024%
(100.000% with Fillers)
Reported timing to dir timingReports
Total CPU time: 1.86 sec
Total Real time: 2.0 sec
Total Memory Usage: 1390.839844 Mbytes
Reset AAE Options


xgb8870@batgirl:Backend
File Edit View Search Terminal Help
Design Boundary: (0.0000, 0.0000) (93.6700, 91.0000)
Error Limit = 1000; Warning Limit = 50
Check all nets
Begin Summary
Found no problems or warnings.
End Summary
End Time: Fri Jun 12 03:09:25 2026
Time Elapsed: 0:00:01.0
******** End: VERIFY CONNECTIVITY ********
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.2 MEM: 0.000M)
innovus 73> verifyProcessAntenna -reportfile mac_array_3x3.antenna.rpt -error 1000
******* START VERIFY ANTENNA ********
Report File: mac_array_3x3.antenna.rpt
LEF Macro File: mac_array_3x3.antenna.lef
Verification Complete: 0 Violations
******* DONE VERIFY ANTENNA ********
(CPU Time: 0:00:00.1 MEM: 0.000M)
innovus 74> report_area
Depth Name #Inst Area (um^2)
0 mac_array_3x3 2751 5827.262
1
innovus 75>


xgb8870@batgirl:Backend
File Edit View Search Terminal Help
Starting SI iteration 2
Glitch Analysis: View an -- Total Number of Nets Skipped = 0.
Glitch Analysis: View an -- Total Number of Nets Analyzed = 3492.
Total number of fetched objects 3492
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 3494, 19.7 percent of the nets selecte
d for SI analysis
End delay calculation. (MEM=1402.43 CPU=0:00:00.3 REAL=0:00:01.0)
innovus 71> verify_drc -report mac_array_3x3.drc.rpt -limit 1000
f-report mac_array_3x3.drc.rpt   # string, default="", user setting
*** Starting Verify DRC (MEM: 1394.4) ***
VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area : 1 of 4
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 2 of 4
VERIFY DRC ..... Sub-Area : 2 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 3 of 4
VERIFY DRC ..... Sub-Area : 3 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 4 of 4
VERIFY DRC ..... Sub-Area : 4 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:00.4 ELAPSED TIME: 1.00 MEM: 30.8M) ***


xgb8870@batgirl:Backend
File Edit View Search Terminal Help
VERIFY DRC ..... Sub-Area : 3 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 4 of 4
VERIFY DRC ..... Sub-Area : 4 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:00.4 ELAPSED TIME: 1.00 MEM: 30.8M) ***
innovus 72> verifyConnectivity -type all -error 1000 -warning 50
VERIFY_CONNECTIVITY use new engine.
******** Start: VERIFY CONNECTIVITY ********
Start Time: Fri Jun 12 03:09:24 2026
Design Name: mac_array_3x3
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (93.6700, 91.0000)
Error Limit = 1000; Warning Limit = 50
Check all nets
Begin Summary
Found no problems or warnings.
End Summary
End Time: Fri Jun 12 03:09:25 2026
Time Elapsed: 0:00:01.0
******** End: VERIFY CONNECTIVITY ********
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.2 MEM: 0.000M)



Path:

/home/xgb8870/Lab3_MAC_Array